

Integrating HEC-RAS with machine learning and deep learning models as a rapid tool for river water level prediction

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Abstract

For flood management, accurate water surface elevation predictions are necessary. hydraulic models such as HEC-RAS produces detailed water surface elevation predictions but requires extensive input data and computational time. This study tested a hybrid approach on the Khorram Abad River in Lorestan Province, Iran, combining HEC-RAS with machine learning and deep learning models. HEC-RAS version 6.0 generated 856 records from 167 cross-sections and eight return periods (2 to 500 years). The dataset was split by cross-section (74 for training, 33 for testing) to prevent information leakage. Four models were trained: Random Forest (RF), Support Vector Machine (SVM), Deep Neural Network (DNN), and Recurrent Neural Network (RNN). RF had the highest accuracy in testing, with an R^2 of 0.9909, an RMSE of 0.619 meters, and an MAE of 0.446 meters. Channel bed elevation accounted for 32% of the prediction importance, and log-transformed discharge accounted for 24%. RF maintained its accuracy across all return periods. Repeated cross-validation gave a mean R^2 of 0.9933 with a standard deviation of 0.0029. The Wilcoxon test showed RF outperformed SVM ($p = 0.000$), DNN, and RNN ($p < 0.001$) with statistical significance. Bootstrap uncertainty analysis produced 95% confidence intervals for RMSE [0.5546, 0.8363] meters and R^2 [0.9834, 0.9927]. The framework provides a faster alternative to HEC-RAS for this river and may be useful for design flood estimation and hazard mapping. Testing on other rivers and under unsteady flow conditions is needed before operational flood warning use.

Keywords Water level prediction · Machine learning · Deep learning · Surrogate model · HEC-RAS · Khorram Abad River

Introduction

Riverine areas are suitable for human habitation, but residents there have historically been vulnerable to flooding from rising water levels¹. Floods are among the most destructive natural disasters, causing significant damage worldwide every year². As climate change intensifies, flood-related damage is increasing³. Accurate prediction of water surface elevation in rivers is a primary method for flood risk assessment. Traditional methods fall into two categories: hydrological (e.g., the Muskingum-Cunge method⁴) and hydraulic (e.g., the HEC RAS method⁵).

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These models have a strong physical basis, but they require substantial input data and considerable computational time⁶. Data-driven approaches have emerged as an alternative⁷. Machine learning and deep learning algorithms can identify complex, nonlinear patterns. Several studies have applied these models to flood prediction. Gao et al.⁸ used an enhanced Random Forest model to estimate rating curves and achieved high accuracy⁸. Sharafkhani et al.⁹ compared four models—MLP, LSTM, Transformer, and LSTNet—for multi-step-ahead water level forecasting at 20 river gauging stations. The LSTNet model outperformed the others under data drift conditions, achieving a median RMSE of 0.00724 for 4-step-ahead predictions, with each step corresponding to 30 minutes⁹.

Kılıçarslan et al.¹⁰ used a machine learning surrogate model to improve flood inundation mapping. They used the HAND FIM method as the baseline and HEC RAS outputs as the target data. The surrogate model reduced false alarms by 18% and improved the Critical Success Index by 26%. The authors note the approach is fast and low-cost, which makes it useful for areas with limited data and resources¹⁰. Yükseler¹¹ compared Random Forest with HEC RAS and found Random Forest had better accuracy¹¹. Duangkhwon et al.¹² developed an integrated deep learning framework with LSTM for water level prediction and CNN for flood inundation mapping. The framework replaces the HEC RAS 1D/2D model. They trained the models on observed hydrological data and HEC RAS outputs. The LSTM predicted water levels with satisfactory accuracy, and the CNN converted those predictions into flood depth maps that matched HEC RAS outputs closely¹². Kordi Karimabadi et al.¹³ used LSTM for streamflow forecasting over a 20-year horizon. They then put the extracted peak discharge values into HEC RAS 2D to produce flood inundation and hazard maps. The LSTM model had R^2 values of 0.98 during training and 0.92 during testing. The study used a 40-year daily time series from the Nesa River in Iran. Estimated reservoir inflow volumes for 25-, 100-, and 500-year return periods were 76.26, 148.73, and 149.22 million cubic meters¹³. Alamu et al.¹⁴ reviewed AI and machine learning surrogate models in Computational Fluid Dynamics. The review covered Reduced Order Models, Physics Informed Neural Networks, Neural Operators, Transformers, and Generative Models¹⁴. Haces Garcia et al.¹⁵ showed that Deep Neural Networks (Feedforward, Bayesian, Physics Informed, and LSTM) work as surrogate models for 2D HEC RAS simulations. The models achieved speedups of 15 to 52 times over conventional models. Median RMSE for cell flooding depths reached as low as 2 mm after fine-tuning¹⁵.

Holmberg et al.¹⁶ developed a recurrent neural operator surrogate for HEC RAS that accelerates simulation speed¹⁶. Feng et al.¹⁷ used a CNN-LSTM-ATT architecture to predict flood depth and inundation extent¹⁷. Salisu et al.¹⁸ reviewed 30 studies and found that machine learning models, especially Random Forest and LSTM, consistently outperform traditional hydrological methods for flood prediction¹⁸. Qu et al.¹⁹ proposed a dynamic framework for chain risk assessment of floods. The framework integrates the HiPIMS hydrodynamic model, a physics-informed deep learning network (PG-ST-MoE), and complex network theory. The authors report the framework is effective for identifying hidden systemic risks in data-scarce regions¹⁹. Lee and Chang²⁰ introduced MC-PINN, a modular coupled physics-informed neural network with a 1D PINN for sewer flow and a 2D PINN for surface flow. The network enforces mass and momentum conservation through physics-based loss functions²⁰. Oluwadare et al.²¹ integrated an ensemble model (Random Forest, SVM, and XGBoost) with HEC RAS 2D and achieved an AUC of 0.979²¹. In their systematic review, Maddodi et al.²² confirmed that machine learning models outperform traditional methods. Most of these studies were conducted in data-rich basins²². Regions with limited data, such as the Khorram Abad River basin, have received less attention²³. The area has consistent flood risk from its topography and torrential rainfall, but has few active hydrometric stations. Recently, hybrid approaches combining physically based hydraulic models with machine learning have gained attention. Poursaeid²⁴ studied water level fluctuations in Miqan Lake in Iran using an optimized SLFFN model with evolutionary algorithms (PSO, GA, ICA) and RROC/PDF uncertainty analysis. That study focused on a lake system. The present research addresses riverine flood prediction, which involves different hydraulic dynamics. Poursaeid²⁴ relied on field observations, while the present study uses HEC RAS simulations as the physics-based data source. This approach enables systematic scenario generation and is useful for data-scarce regions where continuous field records are not available²⁴. Hybrid approaches combining HEC RAS and machine learning have been applied in

other river systems¹⁰. The present study has four specific methodological features: a validation framework with cross-validation, Wilcoxon test, and bootstrap analysis; longitudinal and spatial hydraulic analyses; application to the data-scarce Khorramabad region; and a systematic comparison of four machine learning algorithms with evidence-based model selection. The study predicts water surface elevation using peak flood discharge, channel bed elevation, energy slope, section widths, and Manning's roughness coefficient as inputs. It uses the validation and spatial analysis framework described above.

Materials and methods

Study area introduction

The Khorramabad watershed is in Lorestan Province in western Iran. It covers 2,351 km² and is part of the larger Karkheh River basin, which drains into the Persian Gulf and the Gulf of Oman^{23,25}. The Khorram Abad River starts at the confluence of the Robat and Karganeh rivers in the center of Khorram Abad city and is a major tributary of the Kashkan River. The studied section of the Khorram Abad River runs about 10 km between the Bahramju and Cham Anjir hydrometric stations (Fig. 1).

Research methodology

This applied developmental study aimed to provide a hybrid framework for predicting water surface elevation in the Khorram Abad River. Fig. 2 summarizes the main steps. The framework uses HEC RAS to simulate flow and generate hydraulic data. Machine learning algorithms, particularly Random Forest, are then trained to identify patterns in the data and predict water surface elevation rapidly. The hybrid approach combines the accuracy of the physical model with the computational speed of the data-driven models.

Data collection and preparation

The required data were collected in three categories: hydrological, geometric, and hydraulic. The hydrological data included peak flood discharge for eight return periods from 2 to 500 years (Table 1). The geometric data consisted of 167 cross sections with boundary conditions of a normal slope (0.008 upstream and 0.001 downstream) and a subcritical flow regime. After screening for missing values, outliers, and hydraulic inconsistencies, 60 cross-sections were excluded, leaving 107 cross-sections with complete and reliable data for the final machine learning analysis. Manning's roughness coefficient was set to 0.01 for the main channel and 0.01 for the floodplains, with some cross sections set to 0.015. All data were integrated within a GIS environment. The HEC RAS outputs included water surface elevation, velocity, Froude number, shear stress, and flow power for each cross section and return period. Before feature engineering, the data were screened for outliers and missing values.

Hydraulic simulation and database generation using HEC-RAS

Version 6.0 of the HEC RAS hydraulic model was used for flow simulation and to generate training data. A one-dimensional geometric model was constructed with 167 cross sections, Manning's roughness coefficients, and boundary conditions. Compared to 2D models, 1D HEC RAS is faster²⁶, which allowed generation of a comprehensive training dataset across eight return periods (2 to 500 years). The choice of 1D HEC-RAS is supported by the predominantly longitudinal flow conditions observed in the study reach (Section "[Analysis of Longitudinal Hydraulic Behavior of the River](#)"). Local water authorities also use 1D models for flood forecasting. A total of 856 data records were generated. The main output was water surface elevation at each cross section for each return

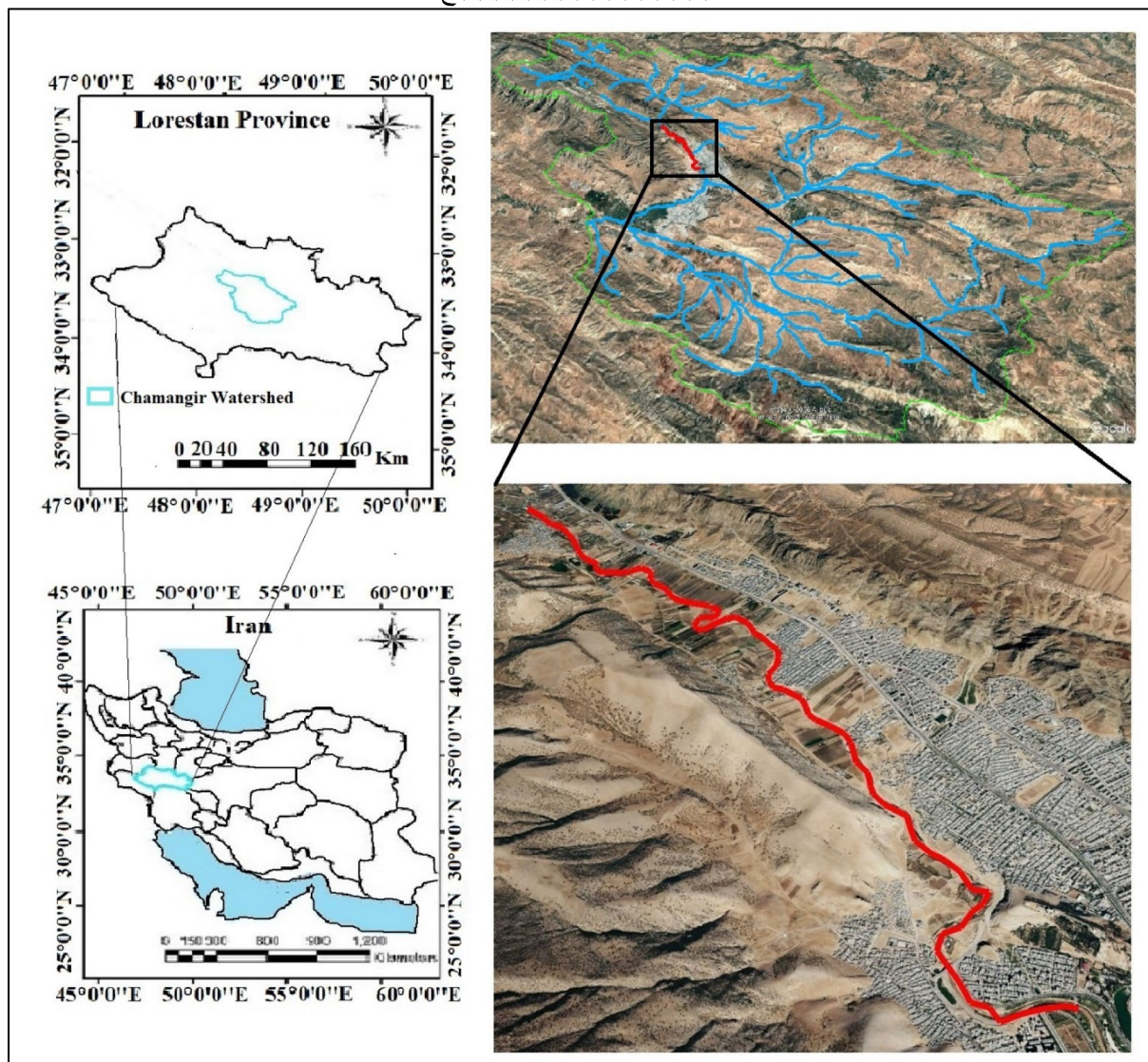


Fig. 1 Geographical Location of the Study Area ***Maps generated using ArcGIS 10.8.1 (<https://www.esri.com>) and Google Earth Pro (Version 7.3.6.10201) [Desktop Software].***

period. Additional hydraulic parameters—discharge, channel bed elevation, energy grade elevation, energy slope, section widths, and roughness coefficients—were extracted and recorded for feature engineering.

Feature engineering and final data preparation

Before normalization, the Shapiro-Wilk test was run on the numerical features. Only 20% of the features followed a normal distribution ($p > 0.05$); most did not. Instead of StandardScaler, which assumes normality, RobustScaler was used. RobustScaler uses the median and interquartile range, which makes it less sensitive to outliers and suited to skewed hydrological data.

Fig. 2 Flowchart of the research steps for predicting water surface elevation in the Khorram Abad River.

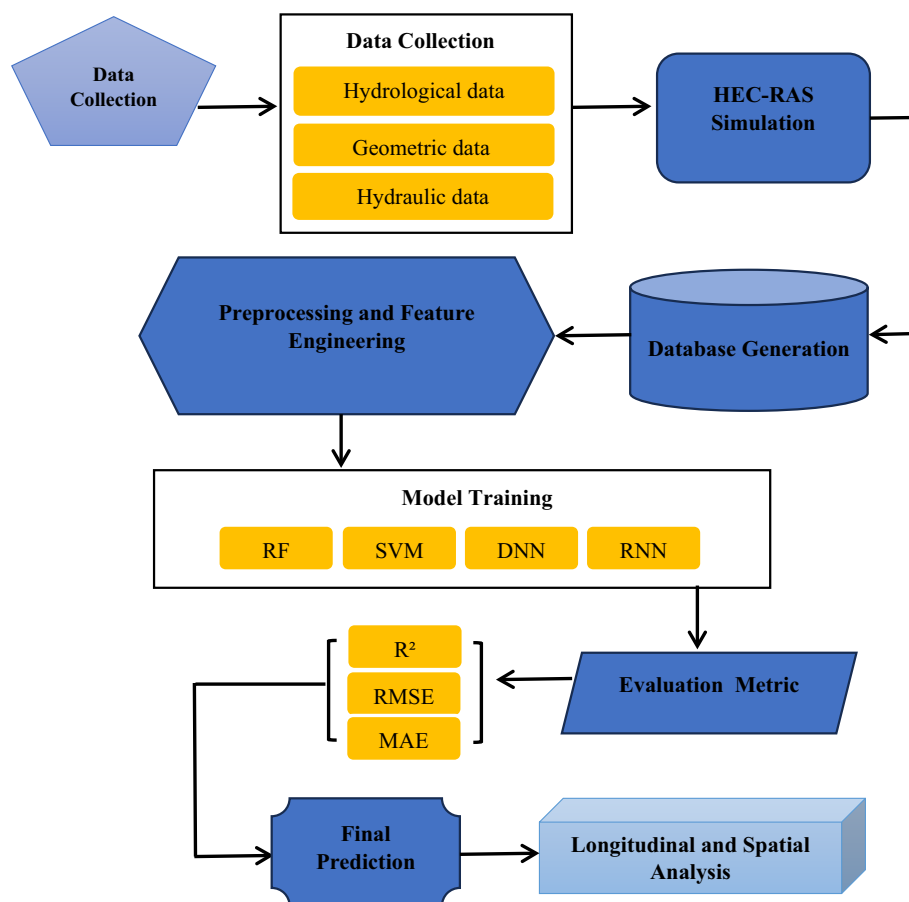


Table 1 Hydraulic parameter values at the cross-section near the Cham Anjir station for different return periods.

Return period	Discharge (m ³ /s)	Water surface elevation (m)	Velocity (m/s)	Froude number	Cross-sectional area (m ²)
T2	49.90	1196.75	3.10	1.00	17.26
T5	80.06	1197.071	3.50	1.00	23.16
T10	127.90	1197.711	3.26	0.79	47.35
T25	225.44	1198.202	3.72	0.80	75.85
T50	313.88	1198.552	4.06	0.81	100.85
T100	417.78	1198.892	4.29	0.80	130.11
T250	573.90	1199.232	4.76	0.84	162.18
T500	704.67	1199.523	4.96	0.84	190.13

Eight new features were created: water depth, the logarithm of discharge, the logarithm of the return period, energy head, the depth-to-energy-head ratio, the slope ratio, total width, and the weighted roughness coefficient. In total, 33 input variables were used.

The dataset was split by cross section rather than random sampling. Seventy-four cross sections (70%) went to training and 33 (30%) to testing. This keeps all eight return periods for each cross section together in one set. The model cannot learn from similar cross sections during training and then be tested on them. This gives a realistic measure of how well the model generalizes to cross sections it has not seen.

Modeling and performance evaluation

Four algorithms were trained to predict water surface elevation: RF, SVM, DNN, and RNN. The RF model used 200 trees with a maximum depth of seven. The SVM model used an RBF kernel with $C = 100$ and $\gamma = \text{"scale"}$. The DNN had three hidden layers with 32, 16, and 8 neurons and a ReLU activation function. The RNN had one recurrent layer with 64 units and a ReLU activation function. Model performance was evaluated using R^2 , RMSE, and MAE.

Longitudinal and spatial analysis

Longitudinal profiles of the channel bed and water surface were plotted for the eight return periods. The water surface gradient was calculated as $S = \Delta H / \Delta L$. A heatmap of water surface elevation as a function of longitudinal position and return period was generated. All analyses used HEC RAS outputs and Python.

Results

Results of hydraulic modeling with HEC-RAS

Running HEC RAS for eight return periods (T2 to T500) across 167 cross sections produced 856 records of hydraulic data. Calibration using data from the Bahramju station gave a Nash-Sutcliffe efficiency of 0.89 for the 25-year return period. As discharge increased, the hydraulic parameters showed systematic variations (Table 1). At the downstream cross section (Cham Anjir), the Froude number went from critical (around 1) for small floods to subcritical (around 0.8) for large floods. The energy slope along the river ranged from 0.0003 to 0.008. For more than 90% of the cross sections and for return periods greater than T10, the Froude number was below 1, indicating subcritical flow under flood conditions.

Quantitative evaluation of predictive models

Four machine learning models were evaluated for predicting water surface elevation: Random Forest, Support Vector Machine, Deep Neural Network, and Recurrent Neural Network. The dataset was split by cross section to prevent data leakage, with 74 sections (70%, 592 samples) used for training and 33 sections (30%, 264 samples) for testing. Model performance was assessed using R^2 , RMSE, and MAE (Table 2). A repeated cross-validation strategy was used to evaluate the Random Forest model's stability. Five-fold cross-validation with 10 repetitions (50 evaluations total) was run on the training data. The mean R^2 was 0.9933 with a standard deviation of 0.0029, and values ranged from 0.9816 to 0.9970. The low standard deviation indicates the model is stable and not sensitive to the training-test split. Random Forest outperformed the other models on the test set, with an R^2 of 0.9909, an RMSE of 0.6192 meters, and an MAE of 0.4464 meters. Training performance was $R^2 = 0.9979$, and test performance was $R^2 = 0.9909$, so the model generalizes well without overfitting. SVM ranked second with an R^2 of 0.9872 and an RMSE of 0.7341 meters. DNN and RNN performed worse, with test R^2 values of 0.7381 and 0.9230, respectively.

Table 2 Comparison of machine learning model performance.

Model	R^2 (Train)	R^2 (Test)	RMSE (Train)	RMSE (Test)	MAE (Train)	MAE (Test)
Random forest	0.9979	0.9909	0.2950	0.6192	0.1816	0.4464
Support vector machine	0.9957	0.9872	0.4175	0.7341	0.3595	0.5480
Deep neural network	0.5313	0.7381	4.3638	3.3194	1.8208	2.4053
Recurrent neural network	0.9613	0.9230	1.2539	1.7993	0.6189	1.3218

Table 3 Results of Wilcoxon signed-rank test for model comparison.

Comparison	p-value	Significant ($p < 0.05$)	Winner
RF vs SVM	0.0000	Yes	RF
RF vs DNN	< 0.001	Yes	RF
RF vs RNN	< 0.001	Yes	RF

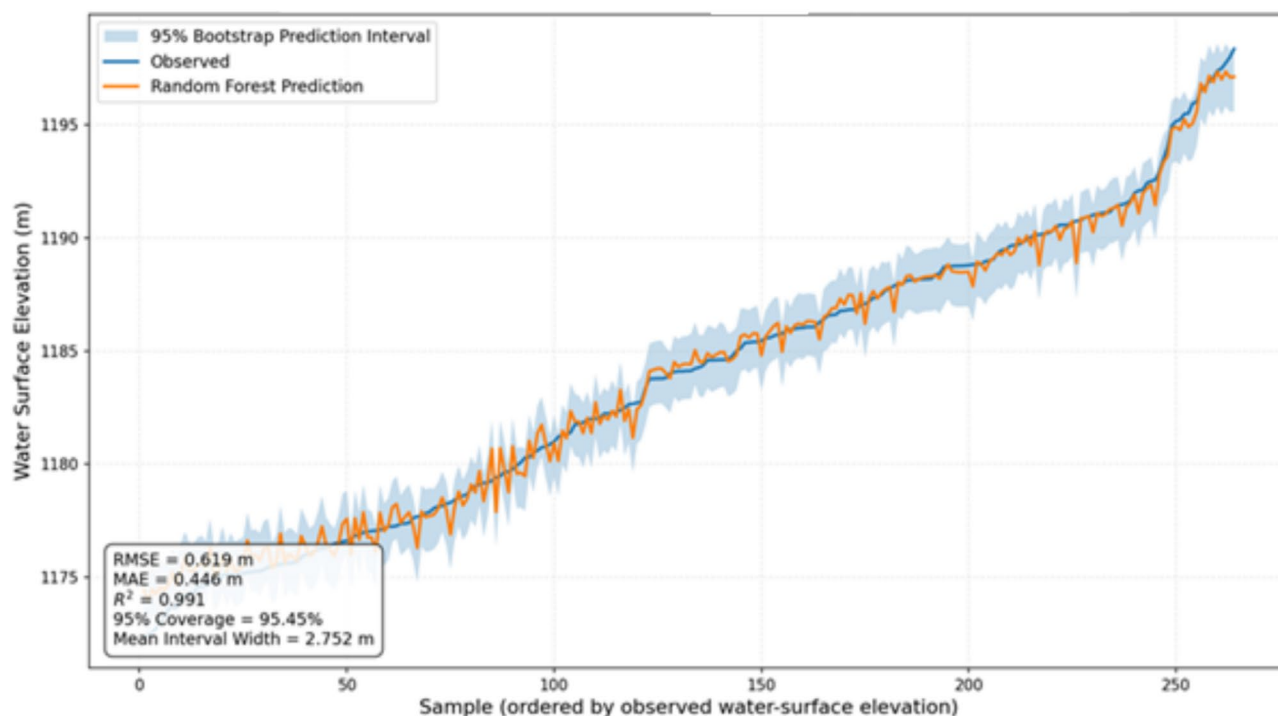


Fig. 3 Predictions with 95% confidence interval (Bootstrap) random forest.

The hyperparameters of the DNN and RNN models were optimized using a manual grid search for a fair comparison. The best DNN configuration had 32 neurons in the first hidden layer with a dropout rate of 0.1 and a learning rate of 0.001. The best RNN configuration had 64 units with a learning rate of 0.005. Both deep learning models still performed worse than Random Forest. The dataset has 856 records, which is small for deep learning. Random Forest is an ensemble method and works well with small to medium datasets. The Wilcoxon signed rank test was run on the absolute prediction errors to check if the performance differences were statistically significant. Random Forest outperformed SVM ($p = 0.0000$), DNN ($p < 0.001$), and RNN ($p < 0.001$). The differences are not due to random chance Table 3.

We ran bootstrap analysis with 1000 iterations on the test dataset to quantify prediction uncertainty. Fig. 3 shows the 95% prediction interval from the bootstrap resampling as a shaded area, with the actual water surface elevations in blue and the mean predictions in red. The prediction intervals give the range where future predictions are expected to fall with 95% probability. The bootstrap results gave a mean RMSE of 0.6790 meters with a 95% confidence interval of [0.5546, 0.8363] meters, and a mean R^2 of 0.9889 with a 95% confidence interval of [0.9834, 0.9927].

Analysis of input variable importance in the optimal model

Variable importance analysis was run on the Random Forest model. Among the 19 input variables, three were most influential. Minimum channel elevation accounted for 35% of the total, making it the most important factor.

The logarithm of discharge was second at 25%. Return period was third at 12%. The remaining 16 features together accounted for 28%.

Validation analysis and error pattern of the best model

The Random Forest model's accuracy was assessed against other models using diagnostic plots. Fig. 4 shows the actual versus predicted values. The two lines are close, with small differences at most points. The model performed satisfactorily on the Khorram Abad River.

Fig. 5 shows the residual plot. The residuals are scattered randomly around the zero line. Most fall within ± 0.5 meters. There is no evident pattern. The random distribution indicates the model is properly trained and has no systematic bias.

Fig. 6 shows the relative feature importance plot. Minimum channel elevation (Min_Ch_Elm) has the highest importance at about 0.32. The logarithm of discharge (Log_Q) is next at around 0.24. Return period, energy head, and water depth are also notable contributors.

Fig. 7 compares R^2 scores across the four models. Random Forest has the highest score, near 1.0 on training data and about 0.99 on testing data. The small gap between training and testing scores indicates overfitting is not a problem for this model.

Fig. 8 shows the RMSE values for the four models on the test data. Random Forest has the lowest error at about 0.61 meters. SVM is next at approximately 0.73 meters. DNN has an RMSE of about 3.31 meters. RNN is not shown in the described results.

Fig. 9 shows the relationship between absolute prediction error and return period. The errors for return periods T2 through T500 fall within a similar range, roughly 0 to 1.5 meters. There is no clear upward or downward trend as the return period increases. The model's accuracy appears consistent across the different return periods.

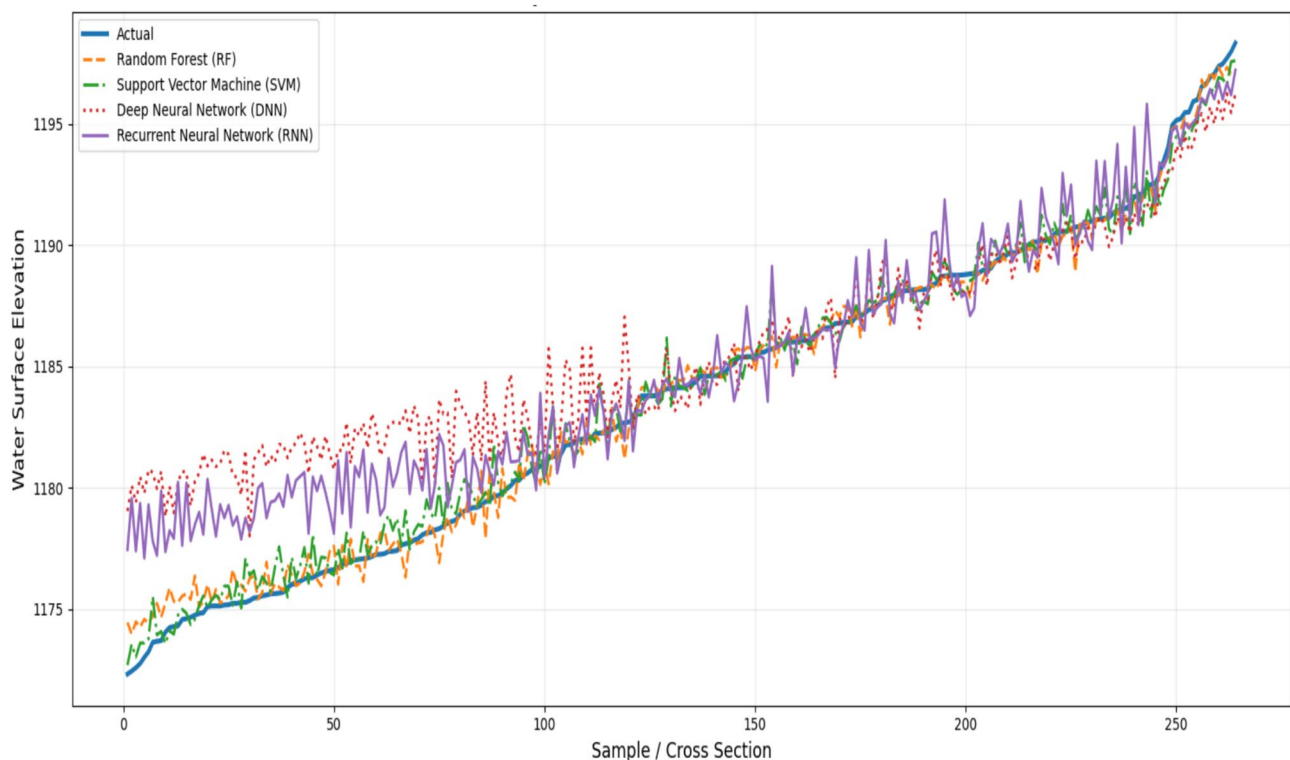


Fig. 4 Comparison of observed and predicted water surface elevation using RF, SVM, DNN, and RNN models in the Khorram Abad River.

Fig. 5 .Residual plot of the random forest model in the Khorram Abad River.

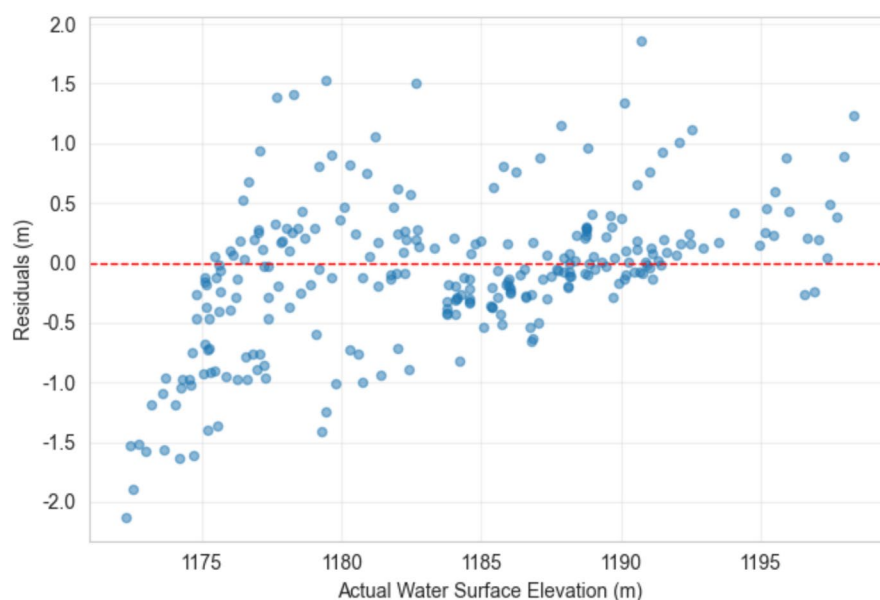


Fig. 6 Relative feature importance in the random forest model for the Khorram River.

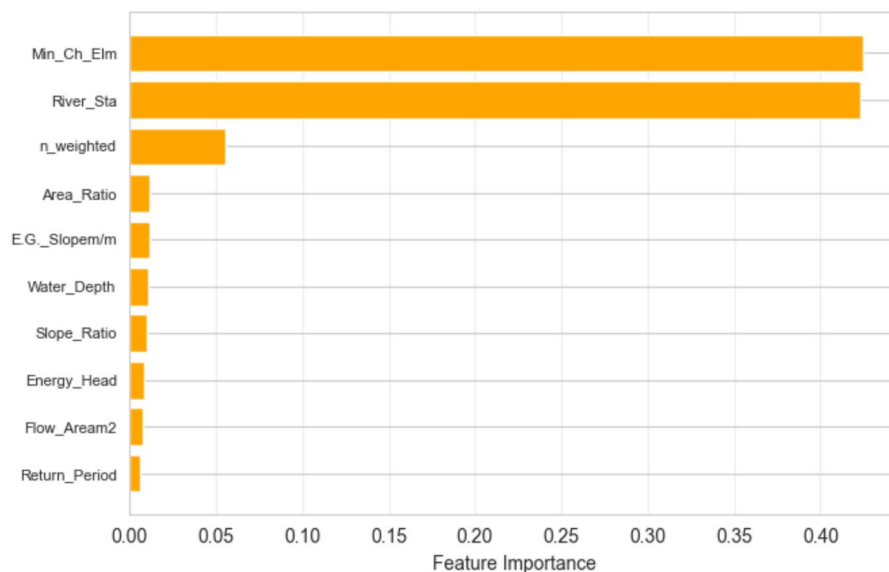


Fig. 10 shows the rating curve for the river reach. As the return period increases from T2 to T500, water surface elevation rises continuously. The actual and predicted values align closely, indicating the model can generate reliable rating curves.

Fig. 11 shows the relationship between absolute prediction error and discharge. The errors are scattered across a wide discharge range, from about 50 to 300 m³/s. There is no clear trend as discharge increases. The model's accuracy does not vary systematically with discharge magnitude.

Fig. 12 shows the distribution of residuals from the Random Forest model. The residuals are centered around zero with a narrow range. The distribution is approximately normal, which suggests the errors are random and the model is properly calibrated. The mean is close to zero, confirming no systematic bias.

Fig. 7 Comparison of the coefficient of determination (R^2) of machine learning models in the Khorram Abad River.

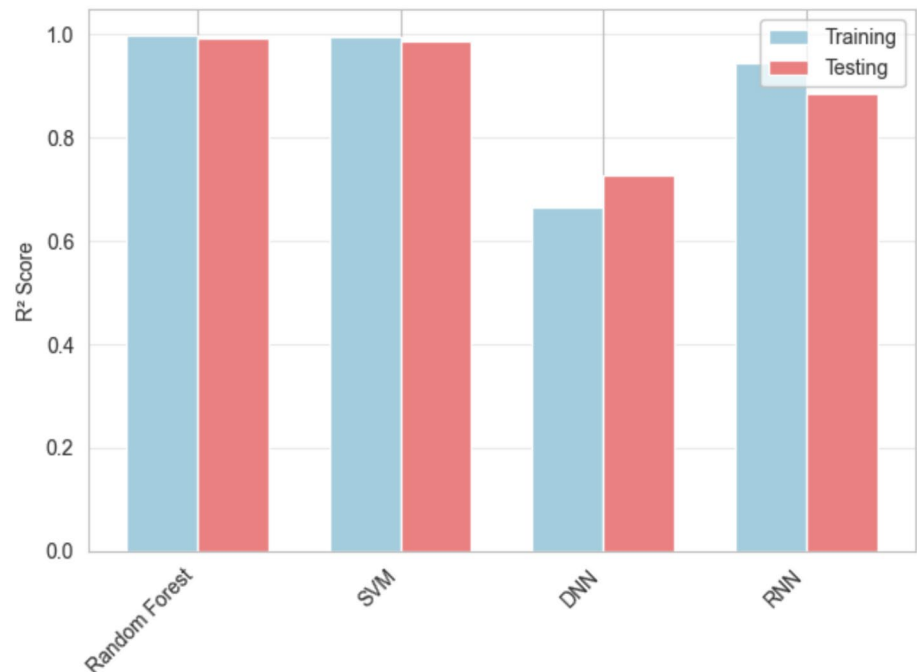
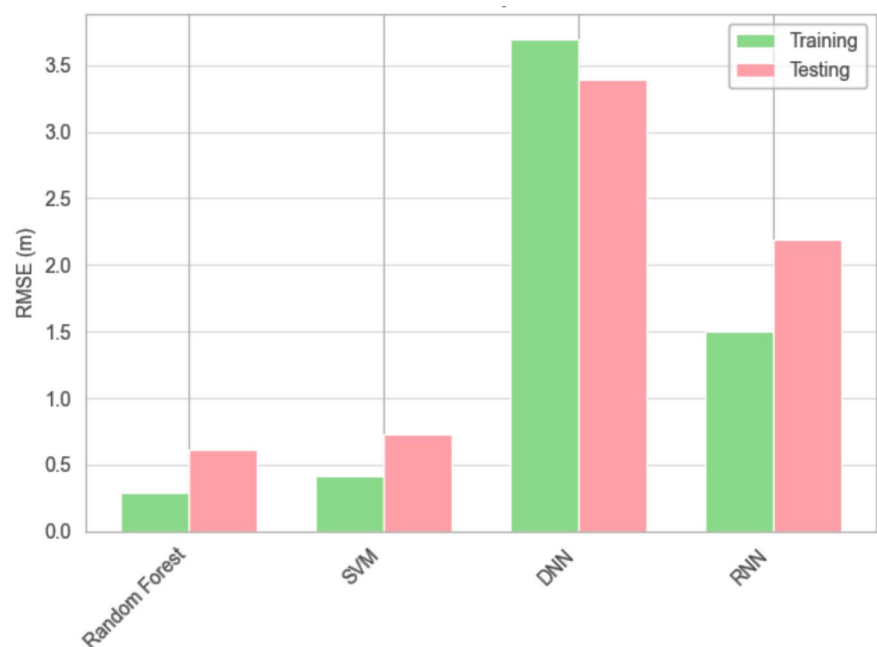


Fig. 8 Comparison of root mean square error of machine learning models in the Khorram Abad River.



Analysis of longitudinal hydraulic behavior of the river

To confirm uniformity of flow conditions and the significance of the observed patterns, statistical tests were used. The Kruskal-Wallis test compared water surface elevations across return periods using all cross-sections together. It found no significant differences ($H = 7.6005$, $p = 0.1074$). The spatial variability in water surface elevation between cross-sections (1167 m to 1198 m) is much larger than the rise from a T2 to a T500 event at a single section (about 3 m). So when all cross-sections are pooled, the return period effect does not show up statistically, even though it is clearly visible at individual cross-sections (Table 1). A one-way ANOVA examined variations in

Fig. 9 Relationship between absolute prediction error and return period in the Khorram Abad River.

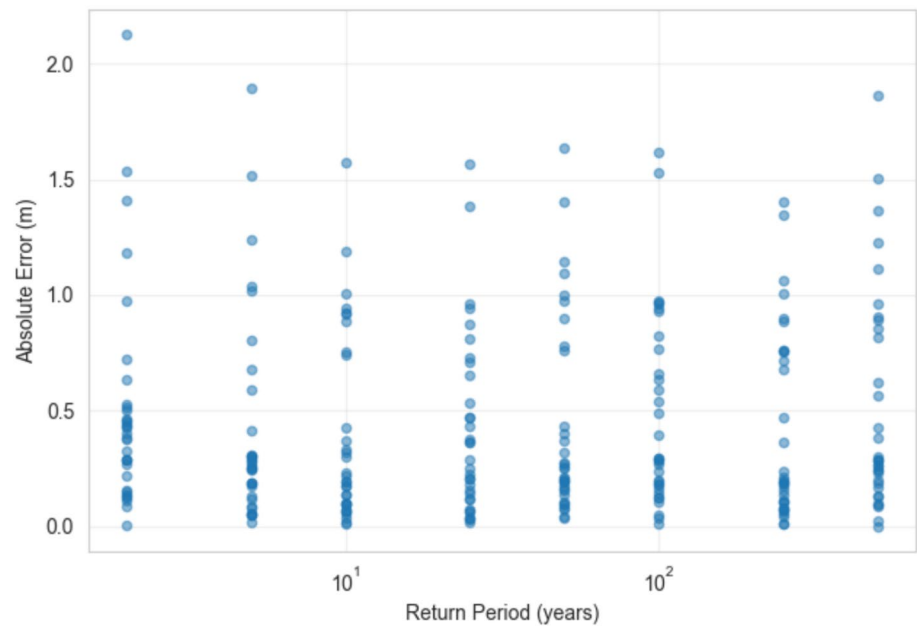
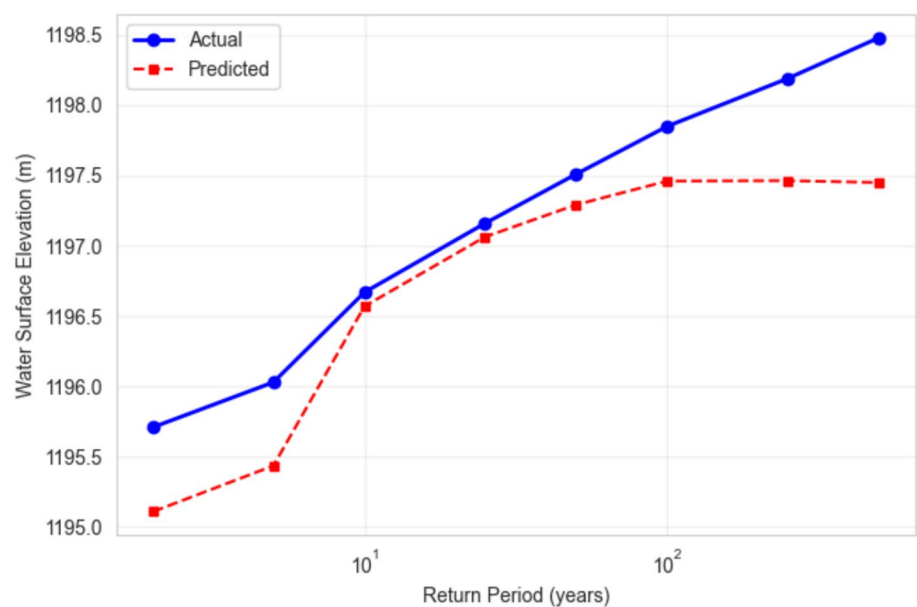


Fig. 10 Discharge-stage rating curve at the index cross-section of the Khorram Abad River.



water surface elevation between cross sections. The results showed significant differences between sections ($F = 16.3114$, $p < 0.001$). The spatial variations are statistically meaningful and not random.

Longitudinal profile of bed and water surface for different return periods

Spatial and longitudinal analyses of the Khorram River study area provided a clear picture of the river's hydraulic behavior. Fig. 13 shows the longitudinal profile. The riverbed has a gentle, nearly linear slope. The water surface lines for all return periods run parallel to the riverbed at a constant distance. The increase in discharge from different return periods has a uniform effect along the entire river length. The system's hydraulic behavior is consistent.

Fig. 11 Scatter of prediction error versus discharge in the Khorram Abad River.

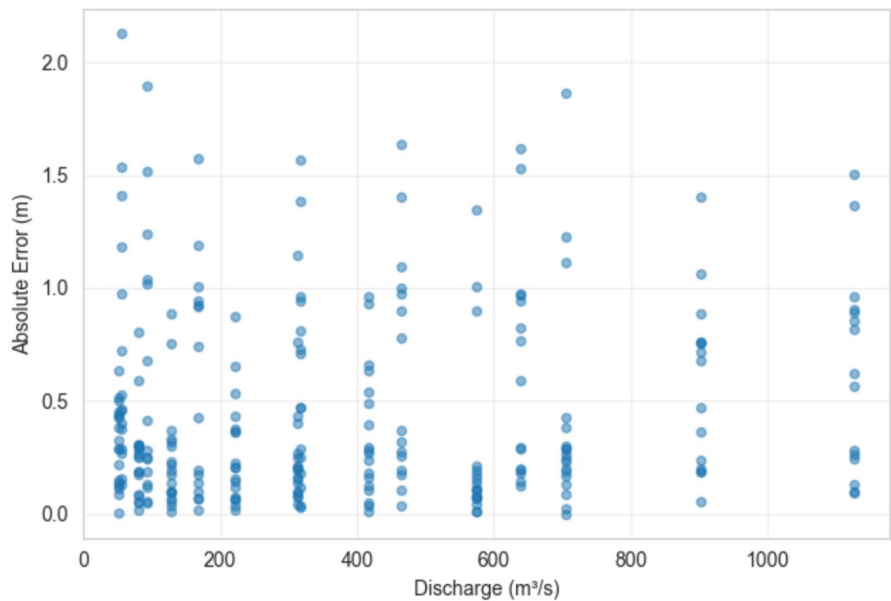
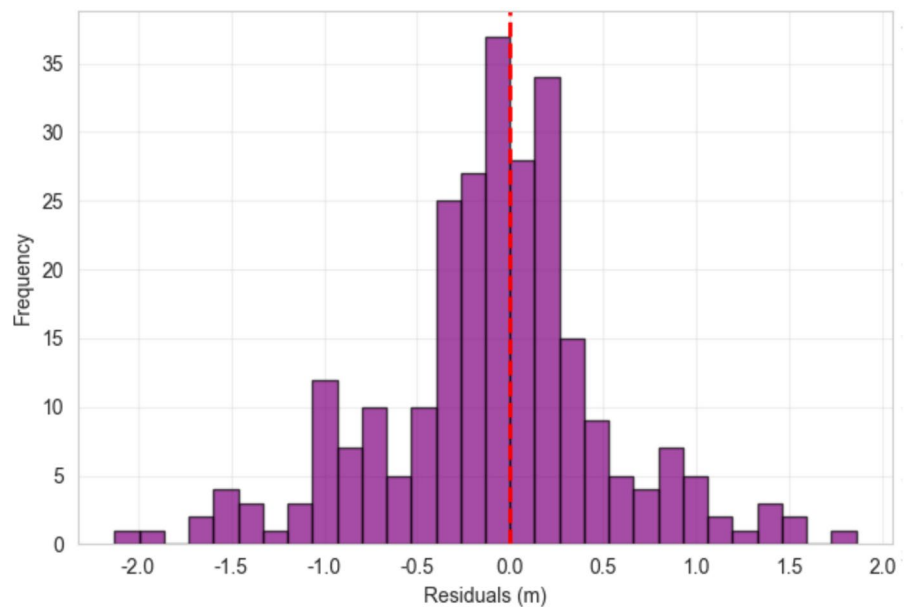


Fig. 12 Frequency distribution of residuals from machine learning models in the Khorram Abad River.



Scatter of water surface elevation as a function of longitudinal position

Fig. 14 shows water surface elevation as a function of longitudinal position (river station). The red regression line has the equation $y = 0.004562x + 1174.191$, indicating a positive linear relationship. The points cluster closely around the line, which demonstrates the accuracy of the relationship.

Water surface variation between cross-sections

Fig. 15 shows the change in water surface elevation between adjacent cross sections for the 100-year return period. Most changes fall between 0.2 and 0.4 meters, with few positive or negative outliers. The water surface rises relatively uniformly along the river.

Fig. 13 Longitudinal profile of bed and water surface in the Khorram Abad River.

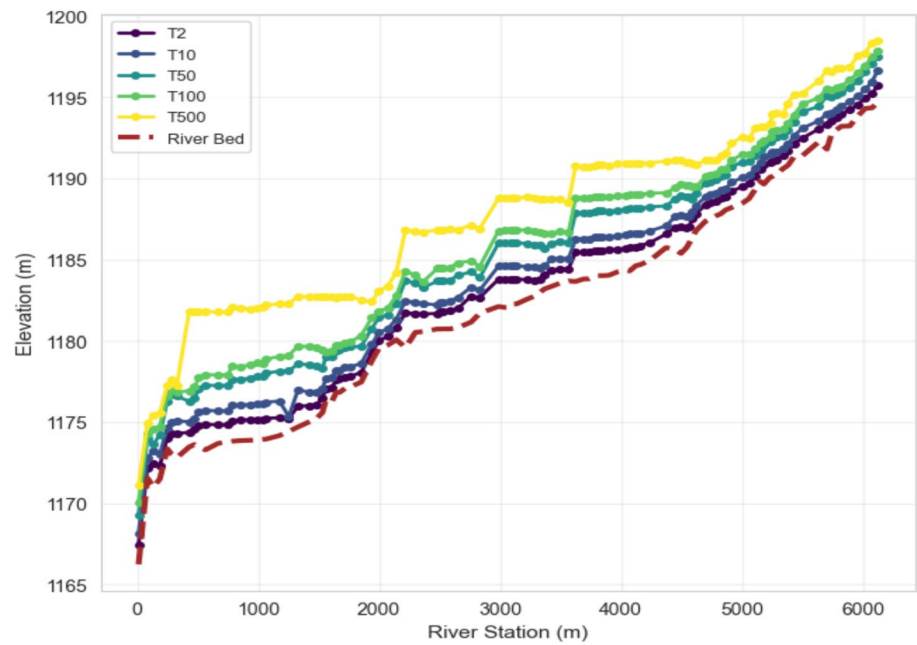
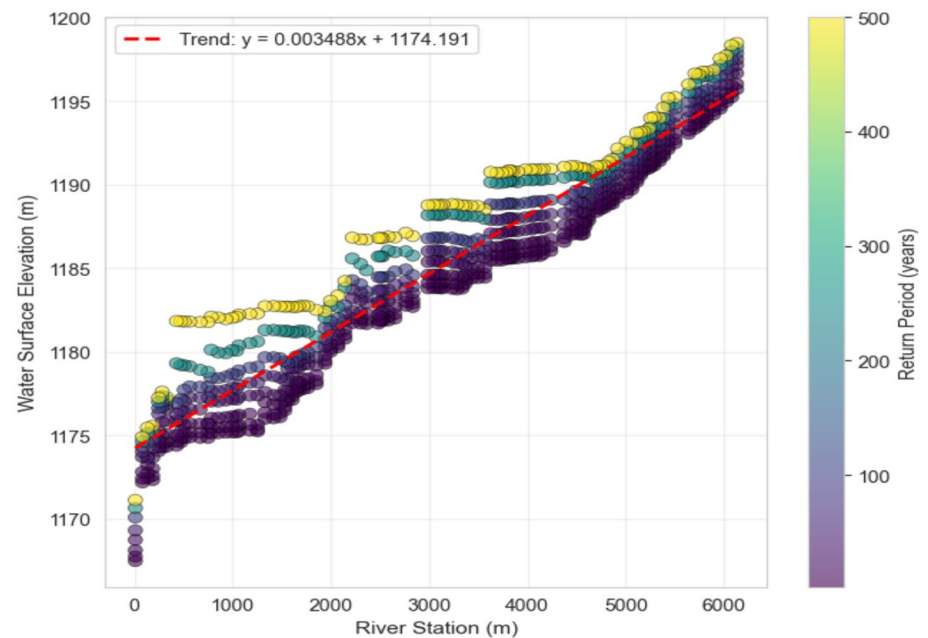


Fig. 14 Scatter of water surface elevation as a function of longitudinal position in the Khorram Abad River.



Water surface gradient along the river

Fig. 16 shows the water surface gradient along the river for various return periods. The lines largely overlap, indicating the gradient is independent of discharge and return period. Gradient values range from -0.01 to $+0.07$ m/m. Negative values occur in some sections, meaning the water surface slope reverses and elevation increases in the flow direction.

Fig. 15 Water surface variation between consecutive cross-sections in the Khorram Abad River (100-year return period).

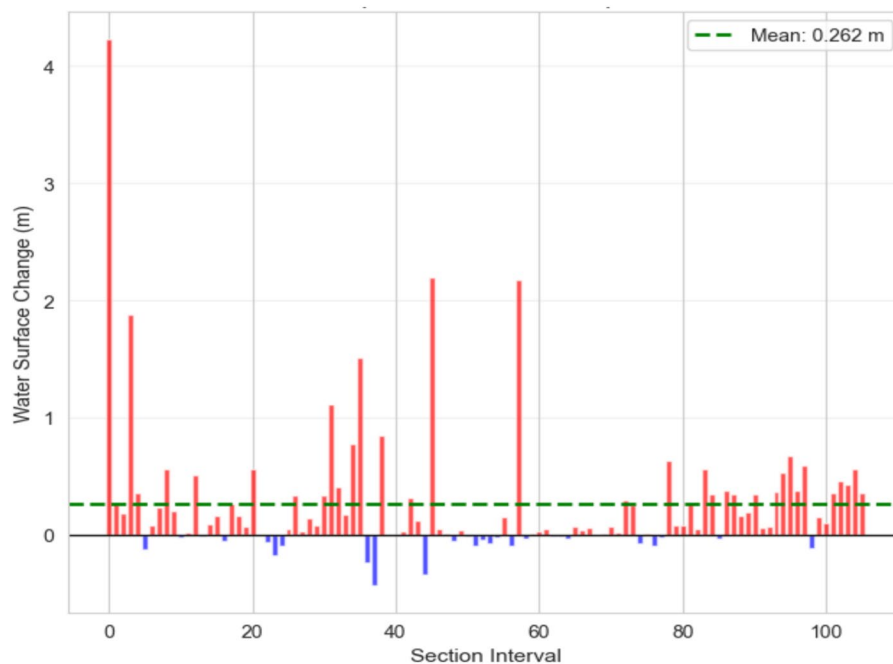
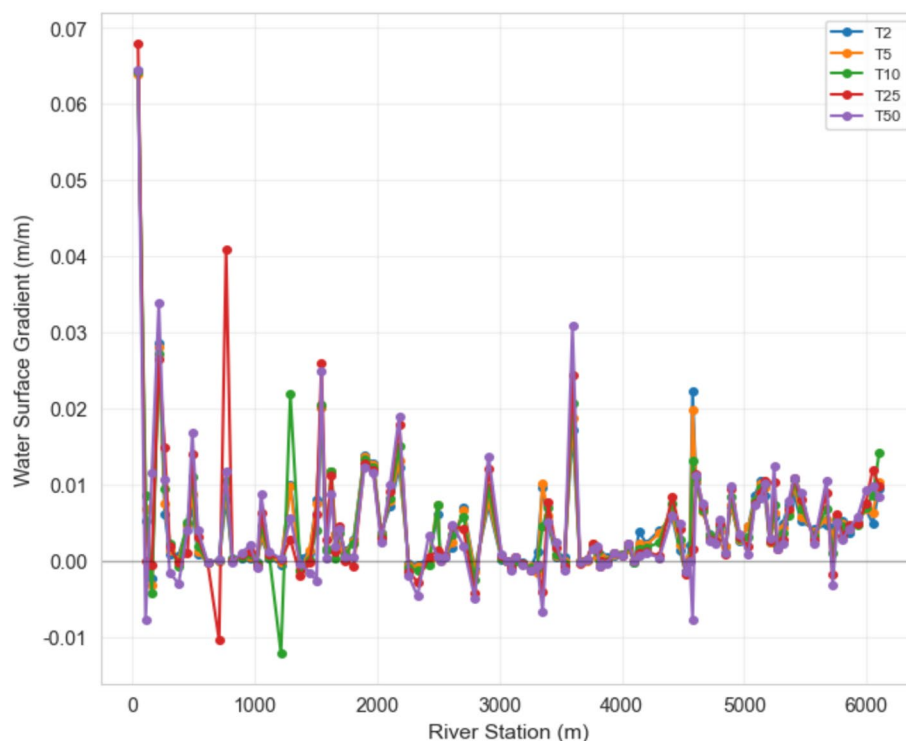


Fig. 16 Water surface gradient along the Khorram Abad River.



Water depth variation along the river

Fig. 17 shows water depth along the river for different return periods. The lines for higher return periods sit at higher levels, indicating greater depths for larger discharges. The lines are nearly horizontal, suggesting water depth remains relatively consistent along the river.

Fig. 17 Water depth variation along the Khorram River.

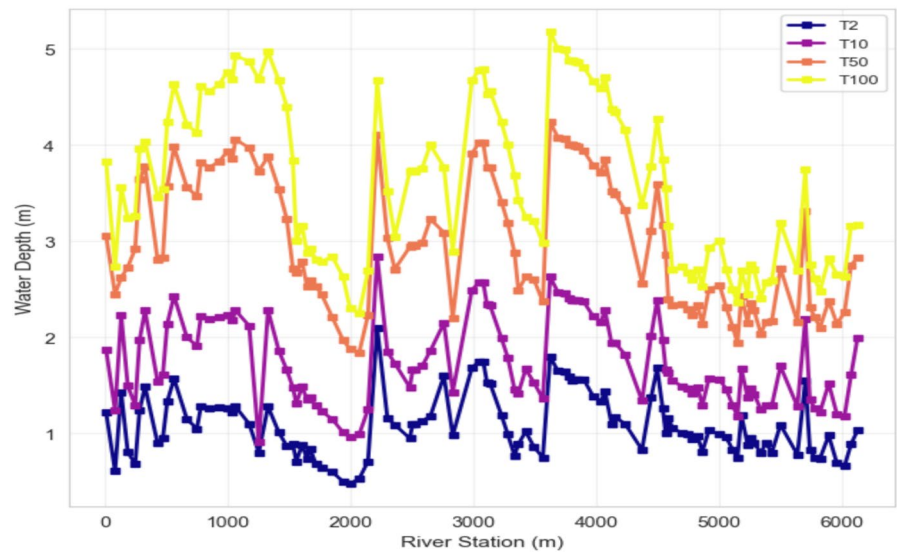


Fig. 18 Heatmap of water surface elevation as a function of longitudinal position and return period in the Khorram Abad River.

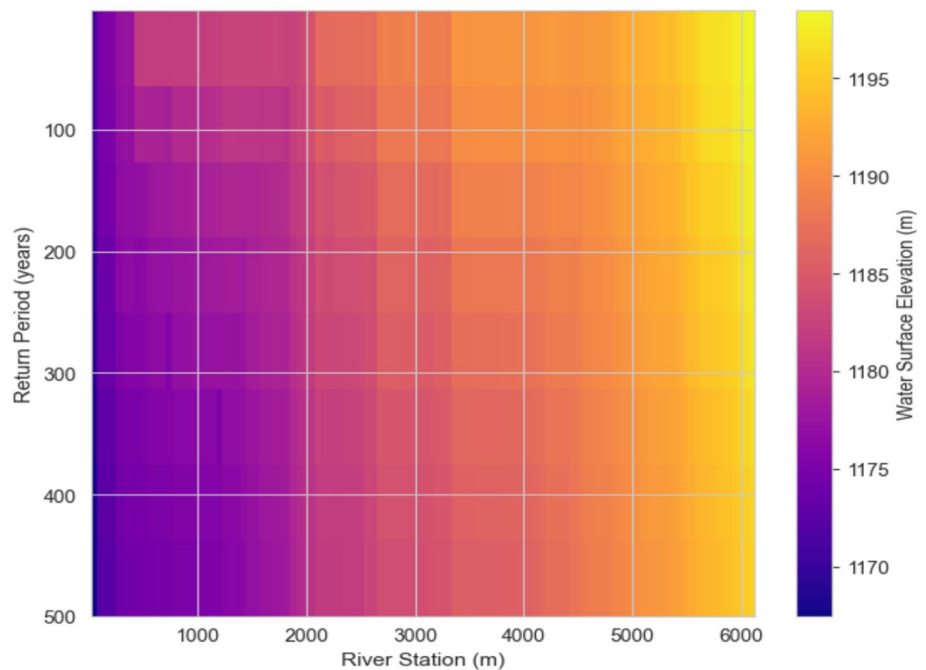


Fig. 18 is a heatmap of water surface elevation as a function of longitudinal position (horizontal axis) and return period (vertical axis). As the return period increases from T2 to T500, the color shifts from purple to yellow, indicating higher water surface elevations. The color change is more pronounced vertically (with return period) than horizontally (with position). Return period has a greater influence on water surface elevation than longitudinal position. For a fixed return period, elevation increases with longitudinal position (moving right), which indicates a positive bed slope and uniform flow along the river.

Discussion

Combining HEC RAS and Random Forest in a hybrid approach can predict water surface elevation in the Khorram Abad River accurately. Physical models like HEC RAS require extensive input data and substantial computational time⁵. This study used HEC RAS outputs as ground truth to train a Random Forest model. The Random Forest model achieved an R^2 of 0.991, with accuracy comparable to the physical model but with near-instantaneous execution, which is useful for rapid flood warning systems.

Zhou et al.²⁷ also found tree-based decision models effective for predicting water levels²⁷. Kılıçarslan et al. (2025) reduced flood inundation prediction error by 18%¹⁰. Feature importance analysis (Fig. 6) showed channel bed elevation (35%) and the logarithm of discharge (25%) were the two most important predictors. This matches fundamental hydraulic principles. The feature importance analysis shows that channel bed elevation and discharge are the main variables controlling water surface elevation in the HEC-RAS simulations. These are the physically meaningful variables one would expect to matter. The model identified them as the most important features. This suggests the surrogate model learned the dominant physical controls in the training data, not spurious correlations.

Regarding data preprocessing, the Shapiro-Wilk test showed that most hydrological parameters (River_Sta, Q_Total_m3/s, Min_Ch_Elm, and W.S._Elevm) do not follow a normal distribution. Hydrological data are often skewed due to the stochastic nature of flood events and variable riverbed topography. RobustScaler was used instead of standard Z-score normalization. Z-score normalization relies on the mean and standard deviation and assumes normality. RobustScaler uses the median and interquartile range, which handles outliers and non-normal distributions better.

Superiority of the hybrid framework and model stability

One finding from this study is the stability of the Random Forest model's accuracy across a range of hydrological conditions. The model produced consistent error rates for both common floods (T10 and T25) and rare floods (T500) (Fig. 9). This has two implications. A model that predicts extreme floods as accurately as moderate ones is useful for crisis management. The stability comes from the coverage of eight return periods in the HEC RAS database and from Random Forest's robustness against overfitting. The Random Forest model also generalized well. Training accuracy was $R^2 = 0.9979$ and test accuracy was $R^2 = 0.9909$. The small difference suggests the model is not tied to a specific training set. A repeated cross-validation strategy was used to check stability. Five-fold cross-validation with 10 repetitions (50 evaluations total) was run on the training data. The mean R^2 was 0.9933 with a standard deviation of 0.0029, and values ranged from 0.9816 to 0.9970. The low standard deviation confirms the model is stable and not sensitive to the training-test split. The cross-validation mean (0.9933) and test performance (0.9909) are consistent, which supports the model's robustness and generalizability. The bootstrap uncertainty analysis supports the Random Forest model's reliability. The 95% prediction intervals in Fig. 3 cover the actual water surface elevations in the test dataset. The model gives reliable predictions with uncertainty that is well characterized. The 95% confidence intervals for the performance metrics were narrow: RMSE between 0.5546 and 0.8363 meters, and R^2 between 0.9834 and 0.9927. These intervals indicate that the model's accuracy is stable. The framework works well for rapid scenario simulation, design flood estimation, and hazard mapping when steady-state assumptions hold. For operational flood warning systems that need real-time hydrograph predictions, the framework would need to be extended to handle unsteady flow conditions. The Wilcoxon signed rank test confirmed the statistical superiority of Random Forest over SVM ($p = 0.0000$), DNN ($p < 0.001$), and RNN ($p < 0.001$). The performance differences are not due to random variation.

The Random Forest model is computationally efficient, with a prediction time of 42.9 milliseconds for 264 test samples (0.1625 ms per sample). This is much faster than HEC RAS simulations and makes the framework practical for operational flood warning. The DNN and RNN models performed well during training but their accuracy dropped during testing (Table 2). This shows the advantage of ensemble methods like Random Forest

for problems with limited data. Deep learning models typically need large datasets, often thousands to millions of samples, to generalize well. With only 856 records, the DNN and RNN models struggled. Random Forest, as a tree-based ensemble, is robust to overfitting and performs well with limited data²⁸. This suggests ensemble methods may be preferable to deep learning for hydrological applications where data are often scarce. Random Forest was selected as the final model for its accuracy ($R^2 = 0.9909$), stability (cross-validation mean $R^2 = 0.9933$, standard deviation = 0.0029), computational efficiency (0.2227 ms per sample), and interpretability through feature importance analysis.

The feature importance analysis

Feature importance analysis showed minimum channel elevation (32%), the logarithm of discharge (24%), and water depth (14%) were the most influential predictors. These results are consistent with the understanding that bed topography and discharge primarily control water surface elevation. Minimum channel elevation represents the bed topography and provides the baseline for all water surface elevations, which makes it the dominant predictor. The logarithm of discharge had a strong influence, not raw discharge. The stage-discharge relationship in open channels follows a power law. For a wide, shallow rectangular channel, the relationship is a power function where the coefficient depends on roughness, width, and energy slope, and the exponent is 5.3²⁹. Taking the logarithm transforms this nonlinear relationship into a linear form, which the Random Forest model can capture more effectively. This follows standard hydrometric practice; logarithmic transformation is a special case of the Box-Cox transformation³⁰, commonly used for calibrating rating curves and producing normally distributed residuals.

Interpretation of river hydraulic behavior and practical implications

The longitudinal analysis showed the flow in the study reach is predominantly uniform and subcritical. This is supported by the parallelism of the water surface lines with the riverbed (Fig. 13) and the consistency of water depth along the river (Fig. 17). At some cross sections, the water surface gradient had negative values, likely from bed constriction or downstream boundary conditions (Fig. 16). Uniform flow simplifies the relationship between discharge and water surface elevation, which is why the machine learning model predicted it accurately. The high importance of channel bed elevation (35%) suggests that accurate cross-section surveying and monitoring of riverbed changes are needed for accurate prediction. Statistical tests on the longitudinal data confirmed that water surface elevation does not vary significantly across return periods (Kruskal-Wallis, $p = 0.1074$), but there are significant differences between cross sections (ANOVA, $p < 0.001$). The hydraulic behavior of the river remains consistent across flood events, while cross-sectional geometry determines local water surface elevations. The framework performed well for the Khorram Abad River case study, but the findings are based on a single river system with 107 cross sections and eight return periods. The practical implications apply primarily to similar river systems. Further studies on diverse river systems are needed to determine broader applicability for flood crisis management and hazard mapping at regional or national scales.

Limitations and future research

1. The training dataset was generated entirely from HEC RAS simulations. The Random Forest model is a surrogate for HEC RAS, not an independent predictor. The approach preserves HEC RAS's physical accuracy but reduces computation time from hours to milliseconds. This makes it useful for rapid scenario-based assessments where steady-state assumptions hold, such as design flood estimation for specific return periods. The model's accuracy depends on the quality and precision of the underlying HEC RAS simulations. Incorporating field measurements and observed hydrological data in future work could make the model more independent and better suited to real-world conditions.

2. The proposed framework was developed and evaluated using a limited dataset of 856 records from a single river system, the Khorram Abad River in Iran. The model showed high accuracy and stability for this case study, but its generalizability to other rivers with different hydraulic characteristics, geometries, and flow regimes has not been validated. The dataset size limits the application of more complex models, especially deep learning approaches, which need larger datasets. Future research should apply the methodology to multiple rivers with diverse conditions—steep slopes, meandering channels, tidally influenced reaches, and floodplain interactions—using larger datasets to assess transferability and robustness.
3. Spatial dimension and flow dimensionality: study used one-dimensional modelling. Extending to two-dimensional (HEC RAS 2D) or three-dimensional models would make the approach applicable to floodplains and urban areas.
4. Temporal dimension and unsteady conditions: The current framework predicts water surface elevation for steady-state conditions. This limits its use for real-time forecasting systems that need continuous hydrograph predictions. The steady-state approach works for design flood estimation and hazard mapping. Future work should extend the framework to unsteady conditions by including temporal dynamics.

Conclusion

This study developed a hybrid framework combining HEC RAS with machine and deep learning models to predict water surface elevation in the Khorram Abad River. Random Forest was the most effective model, with an R^2 of 0.9909 and an RMSE of 0.6192 meters. Feature importance analysis showed channel bed elevation (35%) and the logarithm of discharge (25%) were the most significant predictors. The model maintained accuracy across all return periods (2 to 500 years). Longitudinal and spatial analyses indicated uniform and subcritical flow in the study reach. Integrating HEC RAS with Random Forest combines the accuracy of the physical model ($R^2 \approx 0.9909$) with near-instantaneous predictions (42.9 ms for 264 test samples), a substantial reduction in computational time compared to HEC RAS alone. For the Khorram River case study, the framework shows potential as a quick, cost-effective tool for simulating flood scenarios in similar river systems. Further validation across diverse river systems is needed to establish general applicability. The framework is designed for steady-state conditions, suitable for design flood estimation and hazard mapping. Extending to unsteady flow conditions would be needed for real-time hydrograph prediction in operational warning systems.

Author contributions Ali Haghizadeh: Supervision, Conceptualization, Project administration, Visualization, Validation, Methodology, review & editing, Investigation, Data curation. Leila Ghasemi: Methodology, Software, review & editing, Formal analysis, Data curation. Sanaz vahidimanesh: Writing – original draft, Visualization, Formal analysis, Data curation.

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Data availability Code and output are available. https://github.com/ghasemi9170/Rever_Water_Level.

Declarations

Competing interest The authors declare no competing interests.

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